

ISRO AQI & HCHO - Architecture Summary

This one-page summary captures the core architecture of the project: data sources, processing pipeline, storage, and dashboard.

Key Components

Data sources: Sentinel-5P (TROPOMI), INSAT-3D (MOSDAC), ERA5 (CDS), FIRMS (NASA), CPCB stations

Ingest & ETL: `src/data/*` - download, QC, snap to canonical 0.1 India grid

Feature engineering: `src/features/*` - spatial and temporal features

Models: `src/models/*` - baseline (scikit-learn) and deep (CNN-LSTM / ConvLSTM with PyTorch)

Export: `src/visualization/*` - NetCDF, GeoTIFF, GeoJSON exports for dashboard

Dashboard: `src/webapp/app.py` - Streamlit app reads NetCDF/GeoTIFF and CSV

Optional API: `artifacts/api-server` - Express health + scaffolding

Data Flow (short)

1. Download raw satellite/meteorological/fire/station data
2. Snap to 0.1 India grid and build `grid_daily_features.csv`
3. Train models (baseline or deep) export per-date predictions
4. Export outputs as NetCDF / GeoTIFF / GeoJSON data/processed
5. Streamlit dashboard loads exports and renders maps

Operational Notes

Currently file-based: `data/processed` stores artifacts. Drizzle/Postgres is present for optional metadata storage.

No CI/CD, containerization, or scheduler present. Secrets appear in `.env` (remove from repo).

For production: recommend containerization, object storage (S3/GCS), task scheduler (Airflow/Prefect), and secrets manager.

Contact

Repo: Spatial-ML-Hub - folder: `isro-aqi-hcho`

Generated by automation on request.